



Masaar

Smart tools for Lebanese businesses

Social-commerce operations, cash accountability and decision intelligence

Project promise: Capture the operation, protect the money and stock, explain the business impact, and tell a Lebanese owner what to do next—with evidence.

Project overview

Masaar is the operational system behind Lebanese businesses that sell through Instagram, WhatsApp, TikTok, Facebook, websites, phone or a store. It manages the work after the customer decides to order: products, structured addresses, fulfillment, delivery, payments, cash custody, inventory and the decisions produced from those records.

Problem and solution

Problem	Masaar response
Orders and addresses scattered across chats, screenshots and memory.	One confirmed order, customer profile, standardized Lebanese address and full audit timeline.
Delivered is confused with paid; collected cash may remain with a driver.	Separate delivery/payment states, automatic collection entry, cash custody and controlled reconciliation.
Generic dashboards overstate revenue or profit in mixed USD/LBP reality.	Original-currency truth, approved FX snapshots, cost completeness and explainable decision warnings.
Stock, couriers and customer follow-up live in separate tools.	One role-based workflow linking products, reservations, resources, attempts, returns and customer growth.

Core operating loop

CAPTURE → CONFIRM → FULFILL → DELIVER → COLLECT → RECONCILE → DECIDE

Implemented Solution

A platform the judges can operate, not a static prototype.

Major capabilities

Area	Implemented functional idea
Orders & customers	Quick Order, variants/SKUs, server totals, secure confirmation, duplicate warning, profiles, prior addresses and advisory reliability.
Delivery & driver	Internal drivers, freelancers and companies; zones/fees, manifests, attempts, failure reasons and phone-first assigned stops.
Payments & custody	Cash/Whish/other records, partial/prepaid amounts, proof/reference, holder-specific cash and discrepancy-controlled handover.
Stock & supply	Reservations, movements, receipts, low-stock risk, suppliers/cost history, returns, exchanges and condition-based disposition.
Business intelligence	Revenue/profit/collected separation, product/channel economics, repeat growth, Lebanon area map, delivery Pareto and action-ranked warnings.
Forecast & AI	Governed forecasts, anomaly evidence, scenarios and a grounded assistant that cannot execute consequential changes silently.
Launch controls	Role access, in-app notifications, Lebanese administration organizer, customer tracking, PWA/mobile continuity and honest integration states.

Lebanon-tailored value

- USD/LBP prices and costs with owner-approved exchange-rate snapshots and no invented profit.
- Cash, Whish and OMT-friendly payment records; driver-held-cash accountability.
- All eight governorates, configurable served areas, local address/landmark fields and mixed delivery resources.
- Low-connectivity mobile behavior, supplier/stock exposure and administrative reminders suited to local operating constraints.

Technical architecture

Layer	Technology / responsibility
Experience	React, TypeScript, Vite, Tailwind CSS and Recharts responsive PWA.
Application	Node.js, TypeScript, Fastify and Zod contracts; role and tenant authorization.
Data	Microsoft SQL Server schema and SSMS views; Amazon RDS for SQL Server target.
AWS	Cognito identity, S3 files, CloudWatch observability, WAF, Secrets Manager and AWS Budgets.
Quality	52 automated tests, type checks, lint, production build and responsive owner/employee/driver review.

Challenges, Process & Evidence

The implementation protects operational truth before adding intelligence.

Three hard challenges solved

1. Money truth

Delivered does not prove paid, and paid does not prove cash returned. Masaar uses separate payment events, automatic collection posting, holder-specific custody movements and owner/manager reconciliation with mandatory discrepancy explanations.

2. Mixed Lebanese operating reality

USD/LBP pricing, local payment methods, inconsistent addresses and mixed couriers corrupt generic analytics. Masaar preserves original currency, applies approved FX snapshots, standardizes local address/zone data and exposes missing-data limitations.

3. Intelligence trust

Attractive graphs can combine incompatible money or claim profit without costs. Masaar uses metric contracts, completeness labels, drill-through evidence, deterministic warning thresholds and predictions that remain advisory.

Development process

Nine controlled implementation phases covered product contracts; AWS/design foundations; commerce data; orders; delivery and money; stock and returns; decision intelligence; forecasting/AI hardening; and launch-focused expansion. The final pass added SQL Server migration, role realism, immediate refresh after mutations, error checks, final documentation and release evidence.

User roles

Role	Responsibility
Owner	Full control, setup, reconciliation, analytics and approvals.
Manager	Delegated operational controls and team coordination.
Employee	Order capture, confirmation, preparation, fulfillment, payments and stock tasks.
Driver	Assigned deliveries, attempt outcomes and actual collections on mobile.
Analyst	Read-only evidence and business intelligence.

Team contributions

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Submission Links & Final Readiness

Editable fields remain only where an external URL or team detail is not yet available.

Submission item	Value / status
GitHub repository	Target: https://github.com/shawnlamouza/masaar — local main-branch commit and publishing guide prepared; remote push requires GitHub account access.
Demo application	Local: http://localhost:5173 — public URL requires AWS or hosting account credentials and deployment.
Demo video	[ADD PUBLIC VIDEO URL — maximum two minutes]. Recording script: docs/DEMO_SCRIPT.md.
Functional specification	output/Masaar_Final_Functional_Specification_and_Product_Vision.docx
Implementation plan	output/Masaar_Final_Implementation_and_Release_Plan.docx
User manual	output/Masaar_User_Manual.docx
Database	infra/sql — SQL Server schema, seeds, views and migration resources.
Application archive	output/Masaar_Competition_Submission.zip

Integration truth

- In-app notifications and core local workflows are implemented.
- WhatsApp/email templates support copy-and-send; automated sending requires approved providers, consent and credentials.
- Courier/payment integrations and public AWS deployment require provider/account configuration; the app does not falsely label them live.

Final submission checklist

- Create the empty GitHub repository masaar and run `git push -u origin main`.
- Deploy to AWS/approved host, verify SQL Server/Cognito/secrets/backups and insert the public demo URL.
- Record the two-minute demo and replace the placeholder URL above.
- Add remaining team-member names/contributions, if applicable, then perform the final document review.

Competition close: Masaar is intentionally not another generic ERP. It owns one difficult loop exceptionally well and converts real operational evidence into economically useful decisions for Lebanese businesses.